



Major Project – Abstract

Project Title : Enhancing Arrhythmia Diagnosis Through ECG Deep Learning Classification
Deploying and Augmented Reality 3D Heart Visualization and Interaction
Project Domain : Deep Learning
Project Batch Number : 05
Class and Section :CSD-A

Team Members:

S.No.	Roll Number	Student Name	Mail Id	Contact Number
01	22U11A6724	K.Pranathi	kpranathi0412@gmail.com	7569248273
02	22U11A6716	G.Hyndhavi	hyndhavireddy10@gmail.com	9676149716
03	22U11A6712	D.Yakub	yakubnaikdharavath1309@gmail.com	9392960483
04	22U11A6721	M.Hari venu	Harimiskin07@gmail.com	8317586765

Guide Details:

Name of the Guide	Mrs K. Keerthi Reddy
Designation	Assistant Professor
Department	CSE(Data Science)
Mail Id	kkeerthireddy126@gmail.com
Contact No	7997662648

ABSTRACT

Cardiovascular diseases (CVDs) continue to be a leading cause of mortality globally, highlighting the urgent need for timely and accurate diagnosis. Electrocardiography (ECG) is a vital, manually identifying ECG features and classifying heartbeats is a complex and time-consuming process that demands significant expertise. ArythmiAR offers several key innovations: deep learning-based ECG classification for precise arrhythmia detection, 3D heart modelling and assembly for detailed visualisation, an AR interface for deploying CNN models, 3D localisation of heart sub-regions responsible for arrhythmia anomalies, and enhanced 3D visualisation and interaction capabilities. Our study explores various ECG classification techniques, employing data rebalancing strategies to enhance model performance, with a particular focus on Multilayer Perceptron (MLP) and CNN models, which demonstrated highly competitive results on the PhysioNet MIT-BIH Arrhythmia dataset, achieving an accuracy of 99.07% with the MLP model. Remarkably, this work also involves deploying the ECG classification deep learning model within an AR environment, presenting a prototype for augmented rendering that allows users to localise, visualise, and interact with specific heart regions responsible for arrhythmias. This platform empowers medical professionals to make more accurate diagnoses and develop effective treatment strategies, thereby improving overall patient care.

References:

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Date of Submission :

Team member K Pranathi:

Team member G Hyndhavi:

Team member D Yakub:

Team member M Hari Venu: